

Maven Fuzzy Factory Power BI Dashboard

Project overview

This project analyzes the Maven Fuzzy Factory dataset from Kaggle and turns it into a business dashboard for understanding operations, financial health, and website performance. The report was built on top of an Azure SQL Database pipeline and designed to help stakeholders see how visitors move through the business from landing on the website to placing an order and receiving the shipped items.

Data and model

The dataset was loaded into Azure SQL Database using Python and SQLAlchemy, then connected to Power BI with DirectQuery. The semantic model was shaped to support fast analysis across revenue, traffic, finance, products, channels, devices, and customer behavior.

Dashboard pages

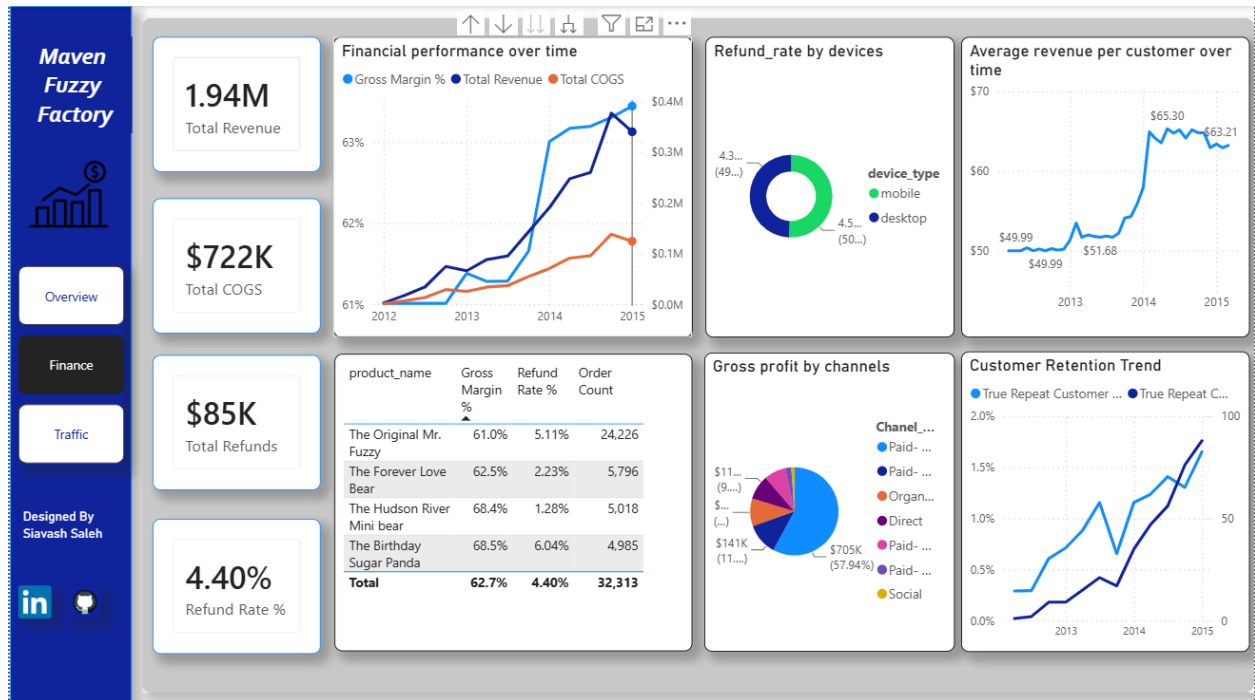
Overview

The Overview page shows the company's core health metrics, including Total Revenue, Conversion Rate, Gross Margin, and Average Order Value. It also breaks down revenue over time, revenue by product, revenue by channel group, and orders by device.



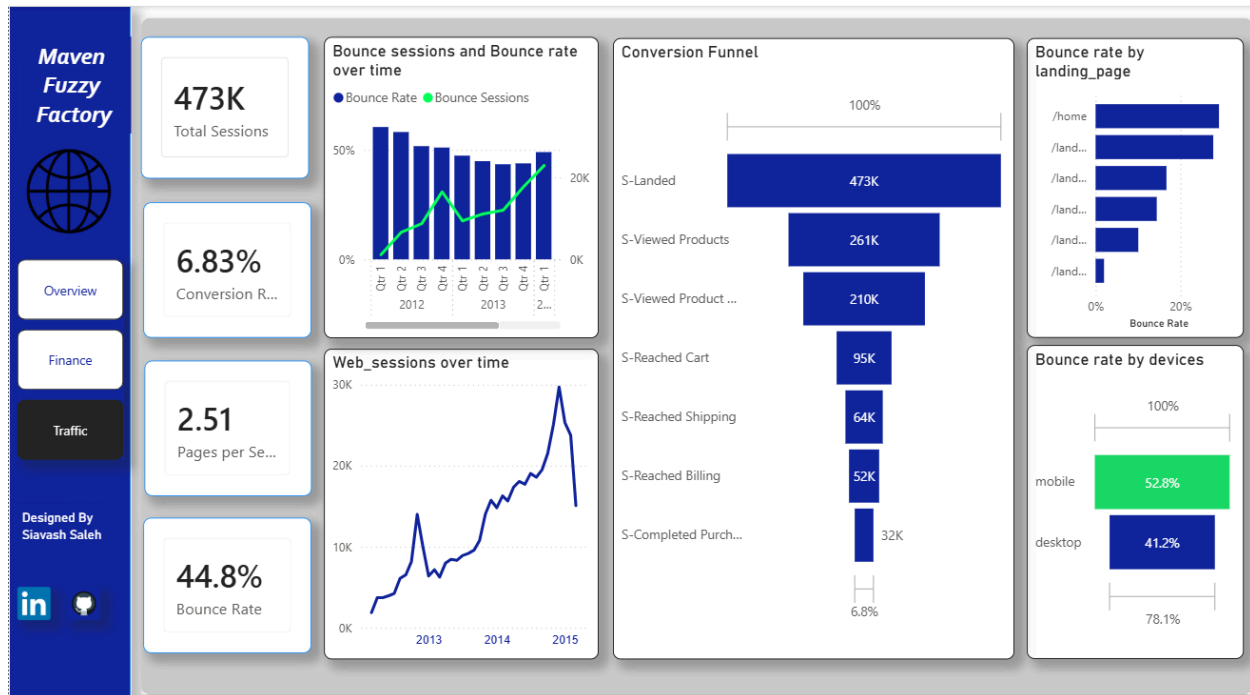
Finance

The Finance page focuses on profitability and customer value. It includes Total Revenue, Total COGS, Total Refunds, Refund Rate, financial performance over time, average revenue per customer, customer retention trend, gross profit by channel, and product-level finance metrics.



Traffic

The Traffic page explains how users behave on the website. It tracks Total Sessions, Conversion Rate, Pages per Session, Bounce Rate, bounce rate over time, the conversion funnel, bounce rate by landing page, and bounce rate by device.



Key findings

The analysis highlights a sharp drop in web sessions after Q4 2014, which may point to a traffic decline or seasonal effect. Lander_4 shows the lowest bounce rate, suggesting it performs best at keeping visitors engaged. The conversion funnel also drops heavily after the product details step, which indicates friction in the purchase journey and a clear opportunity for improvement.

Business value

The dashboard helps the company understand where users enter, where they leave, and where revenue opportunities are being lost. It gives stakeholders a simple way to monitor operations, compare marketing channels, review financial performance, and identify friction points in the customer journey.

Tools used

Python, SQLAlchemy, Azure SQL Database, Power BI, DirectQuery, and Claude via Power BI Modeling MCP Server.